

LIVE

Sepolia → Creditcoin testnet, verified by Attestcoin (Creditcoin USC)

Borrow against your loan book. Without moving it.

Bifrost turns a tokenized loan portfolio into a stablecoin credit line in about fifteen minutes. The portfolio stays escrowed on its home chain; Creditcoin lends against a cryptographic proof of the lock.

■ Sepolia

Collateral escrowed

\$300,000

Block #11,675,618 · never leaves Sepolia

■ Creditcoin

Credit line opened

\$240,000

80% of attested value · USDC

● Attestcoin proof

Demo youtu.be/ufsz9Mp4BbwCode github.com/thewoodfish/bifrost

Live on Sepolia + Creditcoin CC3 testnet

— THE PROBLEM

The loan books live on one chain. The capital lives on another.

■ COMPLIANCE CHAINS

Tokenized, audited loan books

Microfinance funds and invoice factors stay where their KYC and audit stack lives.

The only way across today is a bridge, which adds

✗ a multisig

✗ an oracle

✗ a custodian

■ CREDITCOIN L1

Deep stablecoin liquidity

Credit registries and yield-seeking lenders who want real-world yield.

01 Institutions won't bridge the asset

Wrapping a multi-million-dollar portfolio puts a bridge operator and an oracle in the trust path. That is exactly what institutional risk officers refuse to sign off on.

02 Off-chain credit is slow

A traditional private credit facility takes 3–6 months of legal paperwork before a single dollar moves.

03 So the liquidity stays stranded

Lenders with good collateral can't reach the capital, and capital looking for real yield can't reach them.

 THE INSIGHT

Don't move the asset. Move a proof about it.

The portfolio stays locked in escrow on its home chain. Attestcoin attests the lock, and Creditcoin verifies that proof on-chain before it lends a dollar. Nothing else sits in the trust path.

0

bridges

0

oracles

0

multisigs

1

proof anyone can check

— HOW IT WORKS

Lock. Attest. Lend.



SEPOLIA

Lock on the home chain

- An originator registers the portfolio in `RWAOriginVault`.
- An **independent valuer** prices it. The originator can't price its own collateral.
- The owner locks it against a valuation no older than 7 days, emitting `PortfolioLocked`.

ATTESTCOIN · USC

Attest the block

- USC validators attest the Sepolia block containing the lock, in 8–20 minutes.
- The prover serves a **Merkle inclusion proof** for the transaction.
- Plus a **continuity proof** chaining the block to the attested checkpoint.

CREDITCOIN

Verify and lend

- The pool engine hands the proof to the `BlockProver` precompile.
- It decodes the lock event from the proven receipt and checks every field.
- It opens a USDC credit line at **80% of the attested value**. Draw and repay any time.

— About 15 minutes from lock to funds. The collateral never leaves Sepolia.

USC INTEGRATION

USC isn't a feature here. It's the credit decision.

ON-CHAIN · IN EVERY CREDIT DECISION

CreditcoinPoolEngine

BlockProver.verifyAndEmit 0x...0FD2 GATE

Called inside `attestAndOpenCredit`. **If it doesn't return true, no money moves.**

ChainInfo.get_latest_attestation_height_and_hash 0x...0FD3

Lock freshness enforced on-chain: a lock more than 7,200 blocks behind the attestation frontier is rejected.

BlockProver.verify

A free dry run of the whole claim (`previewIngest`) before the borrower signs, so a bad claim never burns a single-use receipt.

encodedTransaction → **receipt**

Decoded on-chain. Owner, value and valuation round come from the proof, and the receipt hash is consumed on first use.

OFF-CHAIN · SDK AND BROWSER

no backend

@gluwa/usc-sdk · **ProofBuilder**

The CLI fetches proofs through the SDK, retrying the prover's 422 right after attestation lands.

Prover API · **proof-by-tx**

The portal calls the prover straight from the browser, then re-verifies any position live.

ChainInfo.is_height_attested

The signal to start proving. The claim button waits on it, not on a timer.

ChainInfo.get_supported_chains

Read live, and the reason the origin chain is Sepolia. The attestation lag drives every ETA in the app.

— WHAT THE PROOF BINDS

The value comes from the proof. Never from the borrower.

ENCODEDTRANSACTION · PROVEN BY BLOCKPROVER

`abi.encode(uint8 txType, bytes[] chunks)`

LAST CHUNK · THE RECEIPT

`(uint8 status, uint64 gasUsed, logs[], logsBloom)`

PORTFOLIOLOCKED · EMITTED BY OUR VAULT ONLY

`status` ✓ must be 1, or the lock reverted

`owner` ✓ must be the borrower and `msg.sender`

`dollarValue` ✓ sets the credit limit (× 80%)

`valuationRound` ✓ must match the claim

LEARNED FROM THE LIVE NETWORK, NOT THE DOCS

It's an ABI envelope, not receipt RLP

Wrong decoders pass their own fixtures and fail only on real proofs. Ours takes the last chunk, so it works for every transaction type.

Struct field order is load-bearing

Confirmed with differential calls: a transposed `MerkleProof` returns `Unknown selector`.

ChainInfo is snake_case

A camelCase call looks exactly like a missing precompile.

Reverted transactions are still attestable

So the engine checks receipt status before it lends.

 SECURITY MODEL

Try to break it.

Every attack we could think of, and the on-chain check that stops it.



Inflate the collateral value

The value is decoded from the verified receipt, not the claim.

`ClaimDoesNotMatchProof`



Forge or tamper with the receipt

The precompile checks the Merkle and continuity proofs.

`ProofRejected`



Price your own collateral

Originator and valuer are separate vault roles.

`NotValuer`



Lock against an old valuation

Valuations expire after 7 days.

`StaleValuation`



Sit on a proof, borrow later

The lock is aged against the ChainInfo frontier.

`LockTooOld`



Reuse a receipt

Its hash is consumed on first use.

`ReceiptAlreadyUsed`



Use a reverted lock transaction

The receipt status must be 1.

`SourceTxFailed`



Claim someone else's lock

The borrower must be the sender and the proven owner.

`BorrowerMismatch`



Emit from a look-alike contract

Only the configured vault's log counts.

`LockLogNotFound`



Bring a proof from another chain

The chain key is pinned at deploy.

`BadChainKey`

— DON'T TRUST US

Forge a receipt. Creditcoin says no.

Every position has a verify page. Your browser fetches the proof, decodes the receipt and asks the precompile itself. Then it rewrites the dollar value, leaving every other byte untouched.

✓ 1 Lock on Sepolia

✓ 2 Proof fetched

✓ 3 Receipt decoded

✓ 4 BlockProver.verify → true

✓ 5 Receipt unspent

✗ Forged value → rejected

NOW TRY TO LIE

Claim this portfolio is worth more.

Your browser rewrites the dollar value inside the attested receipt — every other byte untouched — and sends it to the same precompile with the same proof. No admin key, oracle or Bifrost server can make this pass.

\$ 7500000

Submit forged receipt✗ **Rejected by Creditcoin in 1989 ms**

Merkle proof validation failed

THE PRODUCT

A complete portal. No backend.

Bifrost

TESTNET

BORROW

Dashboard

LEND

Lend

PROTOCOL

Proof ledger

Valuation desk

Dashboard / Portfolio #2000

Q

Search portfolios...

⌵

0x3656...8DAf

You

Originator

Verify proof

Portfolio #2000

Active line

Registered

Valued

Locked

Attested

Funded

AVAILABLE TO DRAW

\$165,000

of a \$240,000 line · backed by \$300,000 attested collateral

Owed \$75,000 principal + \$0.0439 interest · 8.00% APR, accruing per second

Drawn \$75,000 · Free \$165,000 · 31.3% utilized

Draw

Repay

\$ 0

Draw funds

25%

50%

Max

Up to \$165,000 · wallet holds \$75,000

Collateral

Value\$300,000

Valuation round1

Last valued21 hr 30 min ago

EscrowLocked

Lock tx0xabf9d584...d96d0b ↗

Receipt ID0xe1311dc3d5...

Activity

Repaid \$25,000

Creditcoin 4 min ago tx ↗

Paid \$0.0152 interest

Creditcoin 4 min ago tx ↗

Drew \$100,000

Creditcoin 5 min ago tx ↗

Line of \$240,000 opened against the proof

Creditcoin 5 min ago tx ↗

Block 11,675,618 attested

Attestcoin

SEPOLIA

Collateral locked

Block 11,675,618 · tx ↗

ATTESTCOIN

Block attested

Proof available

CREDITCOIN

Line opened

Verified on-chain

Attestcoin

Live

Lag8 min

Attested#11,681,830

Sepolia head#11,681,868

Active line. Portfolio 2000's real line: interest ticking per second at 8% APR, draw and repay, every step traced back to its chain.

FUNDS READY

\$600,000

Your lock is attested. Creditcoin will verify the proof and release the line in one transaction.

✓ Proof fetched from Attestcoin

Block 11,675,614, tx 97 · 7 Merkle siblings · 87 continuity roots

✓ Verified by Creditcoin's BlockProver

BlockProver.verify() returned true in 192 ms

✓ Claim matches the attested receipt

Owner, value, round and lock age all match the receipt

Claim \$600,000 →

Claim. Proof fetched and dry-run before you sign.

Independent verification

Verify portfolio #2001 yourself

This page doesn't trust Bifrost. Your browser fetches the Attestcoin proof for the Sepolia lock, decodes it, and asks Creditcoin's BlockProver precompile whether it holds — the same check the pool ran before lending a dollar.

Run again

Collateral locked on the origin chain

Transaction0xa551daeb...0e155b ↗

Block11,675,614 · 10 hr 28 min ago

Vault0x13F8...2b7b

Proof fetched from the Attestcoin prover

Attested height11,675,614

Transaction index97

Verify. Anyone re-runs the proof in the browser.

— TRACTION ON TESTNET

Running on live USC proofs today.

■ LOCKED · SEPOLIA

\$300k

Portfolio 2000, valued by an independent valuer

■ PROVEN · USC

\$240k

Credit line opened from the attested receipt

■ DRAWN · CREDITCOIN

\$100k

USDC drawn from the pool from a real wallet

■ REPAID

\$25k

Interest paid first, then principal

78

contract tests, incl. a real attested Sepolia receipt

38

portal tests, fixtures from the live prover

10M

test USDC supplied by lenders for pool shares

0

replays accepted: a reused receipt was rejected on-chain

RWAOriginVault	Sepolia ✓ verified	0x13F8630216EeF192ea74fc2AfA47Bd9edA372b7b
CreditcoinPoolEngine	CC3 testnet	0xBD39e340a43A5693Ae55E0E77b707558Fe67F3e8
BlockProver	CC3 testnet	0x00FD2
ChainInfo	CC3 testnet	0x00FD3
TestUSDC	CC3 testnet	0x6C1e351d926E45Bf88CbdA0412C8831E40AF865B

— WHY IT WINS

Minutes, not months. Proofs, not promises.

	Traditional private credit	Bridged DeFi lending	 Bifrost
TIME TO FUNDS	3–6 months	Minutes	~15 minutes
COLLATERAL LEAVES ITS CHAIN	No, it's on paper	Yes, wrapped	Never
WHAT YOU TRUST	Lawyers and custodians	A bridge multisig	A proof anyone can check
WHO PRICES COLLATERAL	The lender	A price oracle	An independent valuer, on-chain

— BUSINESS MODEL

Asset-light middleware.

Bifrost builds the contracts and the verification pipeline. It never lends its own balance sheet.

RATE SPREAD

● LIVE ON-CHAIN

8% APR paid by borrowers on drawn credit



~6% to lenders, as pool shares grow

~2% protocol reserve (25%)

At full utilization. The pool engine applies the split on every repayment: interest is paid first, and 25% of it goes to protocol reserves.

PLANNED

Origination fee

0.25–0.50% of line value. A \$10M line earns \$25–50k.

PLANNED

Attestation fees

Per cross-chain message, shared with validator incentives.

■ SUPPLY

RWA tokenization platforms

Plume, Centrifuge and Goldfinch already have compliant originators who want cheaper liquidity.

■ DEMAND

Creditcoin capital seeking real yield

Gluwa and the Creditcoin ecosystem route stablecoins into lending backed by real-world loans.

■ FOR CREDITCOIN

Every lock is on-chain activity

Each lock, attestation and draw uses Creditcoin, and it pulls in RWA liquidity stranded on other chains.

WHAT'S NEXT

Each step takes USC deeper.

01

Re-attested valuations

A valuer revalues a locked portfolio, USC proves it to the engine, and the engine shrinks the limit or halts draws. The proof goes from a one-time gate to continuous collateral monitoring.

BUILDABLE ON TODAY'S USC

02

Many origin chains

A vault on each chain and an engine that accepts a set of (chainKey, vault) pairs. Ethereum mainnet is already attested; more follow as USC adds them.

MAINNET READY

03

Proven settlement back

The vault releases escrow on a proof that the line was repaid, removing the last admin key from the lifecycle.

NEEDS RETURN-PATH USC

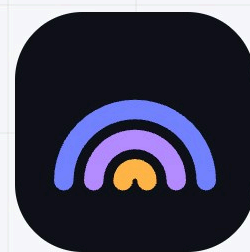
04

Pricing and liquidity

Utilization-based rates, liquidity reserved for undrawn credit, and the origination fee, on top of the pool shares and reserve already live.

ENGINE UPGRADE

Known gaps, stated plainly: • No liquidation yet • Undrawn credit isn't reserved • Escrow release is admin-gated • Unaudited



Borrow against your loan book.

Without moving it.

WATCH THE DEMO

youtu.be/ufsz9Mp4Bbw

READ THE CODE

github.com/thewoodfish/bifrost

BUILT ON

Creditcoin USC · Attestcoin